

Alexandre Sauquet

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EDUCATION

Purdue University
Bachelor's of Science: Computer Science Honors, Artificial Intelligence, Mathematics, Statistics
• Cumulative GPA: 3.88/4.00 | Semester Honors and Dean's List, John Martinson Honors College.
• Relevant Coursework: Data Mining & Machine Learning, Statistical Theory, Artificial Intelligence, Theory of Computation and Computational Complexity (Graduate), Computational Methods in Optimization (Graduate), Elementary Linear Algebra, Linear Algebra II, Numerical Methods, Analysis of Algorithms.

West Lafayette, IN
August 2023 — May 2027

PROFESSIONAL EXPERIENCE

Google | Java
Software Engineering Intern
• Optimizing Workspace monetization by engineering experimentation pipelines to test promotional strategies.

May 2026 — Present
Sunnyvale, CA

CAST Software | FastMCP, Neo4J
Intern AI
• Bridged CAST's code-analysis platform to AI agents in GitHub Copilot and VS Code by developing 44 MCP tool APIs spanning 150+ languages/frameworks, collapsing typical analysis workflows from 20+ UI steps to a single natural-language prompt.
• Built the integration's first deployment path by packaging the MCP server runtime as a Docker container, enabling single-command rollout to enterprise customers.

May 2025 — August 2025
New York City, NY

Duarte, Inc. | HuggingFace, LlamaIndex
Jr. AI Engineer Intern
• Implemented advanced AI models within consulting software solutions by creating custom pipelines, generating empathic human-like presentations that increased production speed by 29%.
• Architected software leveraging large multimodal models that analyzed diverse user-uploaded content with safe guardrails, expanding the company's service capabilities.

May 2024 — August 2024
Santa Clara, CA

Purdue University Research | PyTorch, Optimization Theory
Undergraduate Researcher
• Derived global convergence proofs for gradient descent in the Edge of Stability regime on 2D regularized logistic regression under Prof. Ruizhe Zhang, characterizing dynamics at step sizes beyond the classical stability threshold, with higher-dimensional extensions.
• Built the consultation agent for MAIA, a multi-agent voice assistant expanding rural healthcare access, orchestrating task-specific LLMs in a phone-based pipeline (Twilio speech-to-text, DeepGram voice synthesis) that runs without broadband so patients in low-connectivity areas can reach care by phone.
• Fine-tuned GPT-4o on annotated assembly instruction manuals to localize the current step and highlight the relevant region of each instruction image, delivering direct visual guidance through multi-step assembly from image input alone.

November 2023 — May 2026
West Lafayette, IN

PROJECTS

TE Connectivity AI Cup: 1st place National, AI Assisted Manual Assembly Process Analysis
• Achieved 1st place Nationally and 2nd place Internationally in **TE Connectivity's** Collegiate Competition by outperforming 46 teams from 31 universities.
• Trained and optimized a diffusion transformer model for manufacturing action segmentation by fine-tuning hyperparameters, achieving 88.7% accuracy in production environment analysis.
• Constructed a patented hand detection system using YOLO architecture by implementing automatic task labeling, enabling pipeline generalization across multiple production lines.
• Generated \$170,000 in revenue by securing over 50 deployment opportunities, delivering a complete return on investment within 3 months.

Dec 2023 — May 2024

SKILLS

- Programming Languages:** C++, Python [PyTorch, TensorFlow, Keras, OpenCV, Matplotlib, SKlearn, Pandas, Metalog, NumPy, SciPy, Ultralytics, FastMCP], Java, C, JavaScript, x86_64, Julia, Neo4J, Bash.
- Technologies:** React, Git, GitHub, UNIX/Linux, CVAT, DagsHub, Typst, Latex, SLURM, QuantConnect, LlamaIndex.
- Languages:** French (native), English (bilingual).